Capture and analyze Network traffic using Wireshark.

Aim:

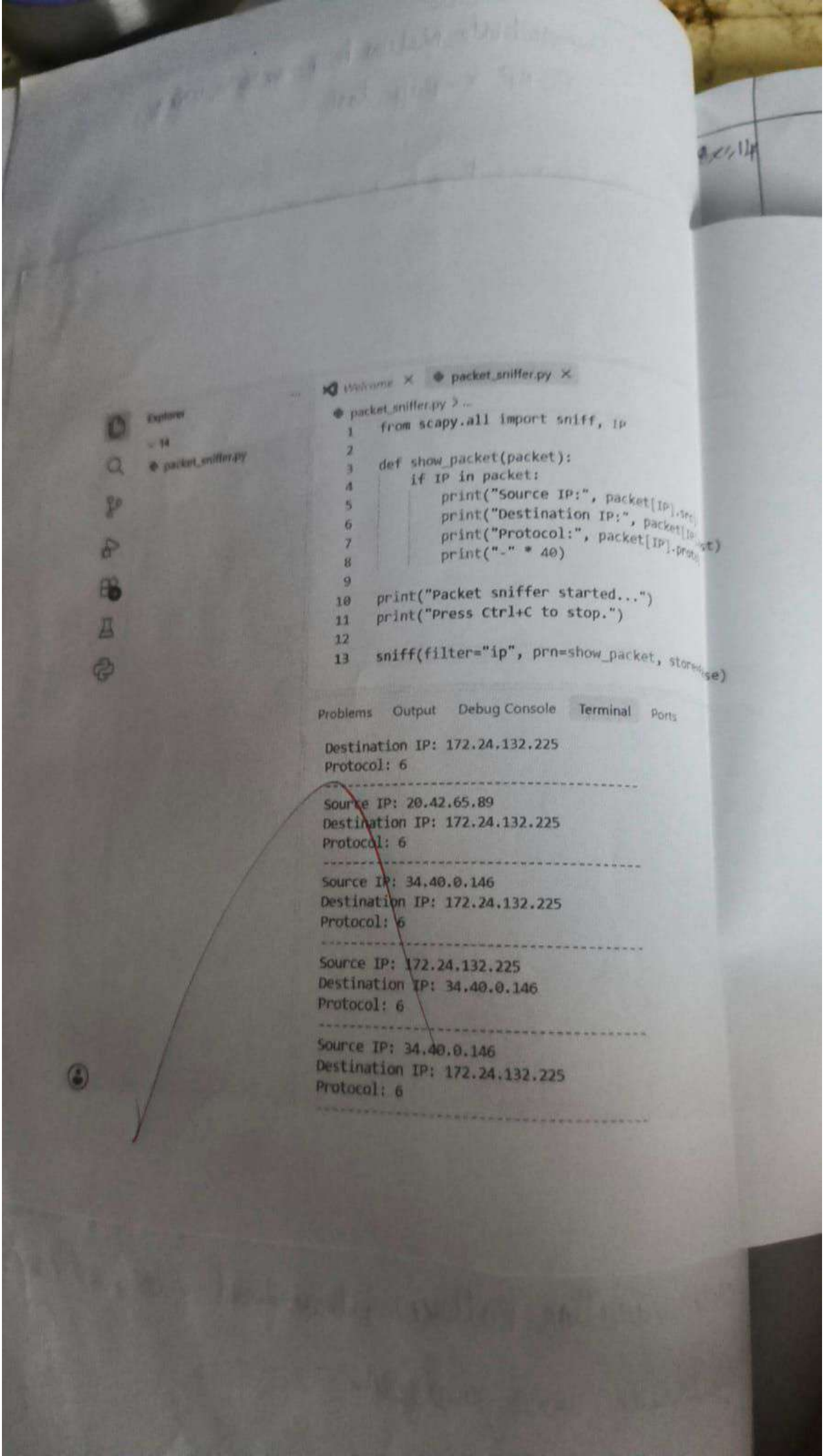
To capture and analyze network traffic using Wireshark and identify the protocols and communication between devices.

Procedure:

* Open Wireshark on the computer.
* Select the network interface to capture traffic.
* Click start Capture to begin capturing packets
* Generate network traffic by browsing networks or using the ping command.
* Observed the captured packets in the Wireshark window.
* Select a packet to examine its protocol details and packet contents.
* Stop the capture and save the captured traffic file if required.

Result:

Thus, network traffic was captured and analyzed successfully using Wireshark, and the packets and their corresponding network protocols were identified.


Explorer
14
packet_sniffer.py
Welcome
packet_sniffer.py
packet_sniffer.py > ...
1 from scapy.all import sniff, IP
2
3 def show_packet(packet):
4 if IP in packet:
5 print("Source IP:", packet[IP].src)
6 print("Destination IP:", packet[IP].dst)
7 print("Protocol:", packet[IP].proto)
8 print("-" * 40)
9
10 print("Packet sniffer started...")
11 print("Press Ctrl+C to stop.")
12
13 sniff(filter="ip", prn=show_packet, store=0)
Problems
Output
Debug Console
Terminal
Ports
Destination IP: 172.24.132.225
Protocol: 6
Source IP: 20.42.65.89
Destination IP: 172.24.132.225
Protocol: 6
Source IP: 34.40.0.146
Destination IP: 172.24.132.225
Protocol: 6
Source IP: 172.24.132.225
Destination IP: 34.40.0.146
Protocol: 6
Source IP: 34.40.0.146
Destination IP: 172.24.132.225
Protocol: 6